

```
import React, { useState } from "react";

function App() {

  const [students, setStudents] = useState(new Map());
  const [filterCourse, setFilterCourse] = useState("");
  const [removeId, setRemoveId] = useState("");

  const [newStudent, setNewStudent] = useState({
    id: "",
    name: "",
    courses: "",
    gpa: ""
  });

  // Add student
  const addStudent = () => {

    const courseSet = new Set(
      newStudent.courses.split(",").map(c =>
c.trim()).filter(Boolean)
    );

    const student = {
      id: Number(newStudent.id),
      name: newStudent.name,
      enrolledCourses: courseSet,
      gpa: Number(newStudent.gpa)
    };

    const updated = new Map(students);
```

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    updated.set(student.id, student);

    setStudents(updated);

    setNewStudent({ id: "", name: "", courses: "", gpa:
"" });
};

// Remove student by ID
const removeStudentById = () => {

    const updated = new Map(students);
    updated.delete(Number(removeId));

    setStudents(updated);
    setRemoveId("");
};

// Convert Map → Array
const studentArray = Array.from(students.values());

// Sort by GPA
const sortedStudents = [...studentArray].sort((a, b)
=> b.gpa - a.gpa);

// Filter by course
const filteredStudents = filterCourse
    ? sortedStudents.filter(s =>
s.enrolledCourses.has(filterCourse))
    : sortedStudents;

// Unique courses
```

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const uniqueCourses = Array.from(
  studentArray.reduce((set, s) => {
    s.enrolledCourses.forEach(c => set.add(c));
    return set;
  }, new Set())
);

return (
  <div style={{ padding: "20px" }}>

    <h2>Course Enrollment Dashboard</h2>

    <h3>Add Student</h3>

    <input
      placeholder="ID"
      value={newStudent.id}
      onChange={e => setNewStudent({ ...newStudent,
id: e.target.value })}
    />

    <input
      placeholder="Name"
      value={newStudent.name}
      onChange={e => setNewStudent({ ...newStudent,
name: e.target.value })}
    />

    <input
      placeholder="Courses (comma separated)"
      value={newStudent.courses}
```

```
        onChange={e => setNewStudent({ ...newStudent,
courses: e.target.value })}
    />

    <input
        placeholder="GPA"
        value={newStudent.gpa}
        onChange={e => setNewStudent({ ...newStudent,
gpa: e.target.value })}
    />

    <button onClick={addStudent}>Add Student</button>

    <h3>Remove Student by ID</h3>

    <input
        placeholder="Enter ID"
        value={removeId}
        onChange={e => setRemoveId(e.target.value)}
    />

    <button
onClick={removeStudentById}>Remove</button>

    <h3>Filter by Course</h3>

    <input
        placeholder="Course"
        value={filterCourse}
        onChange={e => setFilterCourse(e.target.value)}
```

```

/>

<h3>Unique Courses</h3>

<ul>
  {uniqueCourses.map((course, index) => (
    <li key={index}>{course}</li>
  ))}
</ul>

<h3>Students (Sorted by GPA)</h3>

<ul>
  {filteredStudents.map(student => (
    <li key={student.id}>
      ID: {student.id} | Name: {student.name} |
GPA: {student.gpa} | Courses:
{[...student.enrolledCourses].join(", ")}
    </li>
  ))}
</ul>

</div>
);
}

export default App;

```



Course Enrollment Dashboard

Add Student

ID	Name	Courses (comma separated)	GPA	Add Student
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Remove Student by ID

Enter ID	Remove
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Filter by Course

Course

Unique Courses

- DAA
- AP
- economics
- english

Students (Sorted by GPA)

- ID: 87 | Name: rohan | GPA: 9 | Courses: economics, english
- ID: 83 | Name: rahul | GPA: 8.5 | Courses: DAA, AP

Activate Windows
Go to Settings to activate Windows.